RESUME

Gregor Nottmeier

M.Sc. Data Science Student | Applied AI & Reinforcement Learning
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GitHub: https://github.com/gregornottmeier | LinkedIn: https://www.linkedin.com/in/gregor-nottmeier-265734205
Languages: German (Native/C2), English (Fluent/C1), Slovak (Fluent/C1), Spanish (Basic/A1-A2), Latin (Latinum)

PROFESSIONAL SUMMARY

M.Sc. Data Science student at the University of Mannheim with a strong mathematical foundation (B.Sc. Mathematics in Business and Economics) and hands-on experience in reinforcement learning, statistical modeling, and full-stack ML architectures. Proven track record in developing modular data pipelines, fine-tuning transformer models, and automating analytics workflows. Eager to leverage strong algorithm design and object-oriented programming skills to drive next-generation supply chain simulation and industrial optimization at adago.

TECH-STACK & SKILLS

Programming Languages: Python (OOP), SQL, TypeScript, MATLAB, VBA, SAS, LaTeX

Machine Learning & Frameworks: PyTorch, Hugging Face Transformers, Stable-Baselines3, Scikit-Learn, Gymnasium, Braindecode, GeoPandas, OSMnx

Backend & Infrastructure: Docker, PostgreSQL, Flask, Next.js, Nginx, Gunicorn, systemd, Prisma, Linux, Git, GitLab, GitHub, Jupyter

Core Competencies: Reinforcement Learning & Bandit Algorithms, Deep Learning, Large Language Models (LLMs), Statistical Modeling, Time Series Analysis, Simulation & Optimization, Full-Stack ML Deployment

SELECTED PROJECTS

LLM Sentiment Alignment via RLHF

Tilburg University | Python, PyTorch, Transformers, Stable-Baselines3, Gymnasium

Enhanced model alignment stability and mitigated reward-hacking by systematically tuning KL-divergence penalties and adaptive batch sizes during GPT-2 fine-tuning.

Improved positive sentiment generation scores while preserving linguistic diversity by architecting an end-to-end Deep Reinforcement Learning pipeline.

Interactive Surveys Web-App & ML Analysis Pipeline

Team Project (M.Sc. Data Science) | Flask, PostgreSQL, Nginx, Gunicorn, systemd, GitLab

Built a high-performance research platform for analyzing Human-AI interaction dynamics by engineering an asynchronous Flask backend connected to raw PostgreSQL database layers.

Streamlined downstream machine learning evaluation by orchestrating modular background subprocesses and deploying robust production infrastructure under Linux and Nginx.

Contextual Linear Bandit Reinforcement Learning (linUCB / OFUL)

Bachelor Thesis (Universität Mannheim) | Python, Reinforcement Learning, Monte Carlo Methods

Evaluated theoretical regret bounds and algorithm efficiency across dynamic multi-armed bandit settings by implementing custom linUCB and OFUL simulation environments in Python.

Achieved top academic distinction (Grade: 1.3) by rigorously bridging mathematical optimization theory with stochastic empirical modeling.

PROFESSIONAL EXPERIENCE

Mercedes-Benz Bank AG — Intern Risk Analytics Credit

10.2024 – 02.2025 | SAS, Excel (VBA), SAP Bank Analyzer, Power BI

Accelerated month-end closing processes and reduced manual execution time by developing automated VBA macros and SAS data aggregation workflows.

Enhanced credit risk monitoring by designing interactive Power BI dashboards to track Cost of Risk metrics and portfolio exposure.

Integra e.V. (Student Consulting) — Internal Finance Manager & Project Lead

03.2021 – 02.2023 | Project Management, MS Office, Market Analysis

Delivered strategic insights for enterprise clients by co-directing comprehensive B2B customer experience market research initiatives.

Ensured organizational financial stability and operational alignment by managing budget allocation, workshops, and internal strategy projects.

C4 Steuerstrategen GmbH — Working Student

07.2022 – 09.2022 | Data Management, Process Optimization

Maintained 100% data compliance and audit readiness by structuring, verifying, and preprocessing real estate tax assessment records.

Lerne Smart UG / J.C.C. Bruns — Content Creator & Digital Educator

02.2019 – 09.2020 | Digital Education, Agile Execution

Increased digital literacy across target groups by designing and delivering structured, interactive training modules for mobile and desktop systems.

EDUCATION

Universität Mannheim — M.Sc. Data Science
09.2024 – Present | Current GPA: 1.9 (German Scale)

Focus: Machine Learning, Reinforcement Learning, Natural Language Processing, Distributed Computing.

Tilburg University — Erasmus+ Exchange Semester (Data Science / AI)
01.2026 – 06.2026 | Grade: 8.6 / 10 (Excellent)

Focus: Advanced Deep Reinforcement Learning, LLM Alignment, NLP Architectures.

Universität Mannheim — B.Sc. Mathematics in Business and Economics
09.2020 – 09.2024 | Final Grade: 2.2 | Bachelor Thesis Grade: 1.3

Thesis: *The linUCB / OFUL Algorithm in Contextual Linear Bandit Reinforcement Learning*.

Universidad Autónoma de Madrid — Erasmus+ Exchange Semester
09.2023 – 01.2024 | Mathematics & Economic Modeling

Ratsgymnasium Minden — Allgemeine Hochschulreife (Abitur)
Graduated 06.2020 | Final Grade: 1.2

VOLUNTEERING & ENGAGEMENT

Member of Student Council (Fachschaft Math/Info), Universität Mannheim (*Until 10.2024*): Represented student body interests at faculty board meetings and organized technical networking events.

Volunteer Enumerator / Field Interviewer, Zensus 2022 (City of Mannheim) (*05.2022 – 10.2022*): Executed structured household survey interviews adhering to strict data quality and privacy standards.